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Place Your Bets? The Value of Investment Research on Reddit's Wallstreetbets

Contents

1	Big picture	2
2	Data and measurement	2
2.1	WSB DD reports	2
2.2	Pre-GME and post-GME periods	3
2.3	Main social-media variables	3
2.4	Post characteristics: price pressure and attention	3
2.5	Return, cash-flow-news, and trading outcomes	3
3	Models and empirical specifications	4
3.1	Determinants of WSB coverage	4
3.2	Future-return regression	4
3.3	Cash-flow-news regression	4
3.4	Report-type regressions	4
3.5	Retail trading regressions	5
4	Identification and empirical design	5
4.1	What the paper identifies	5
4.2	Controls and benchmarks	5
4.3	Mechanism tests	5
4.4	Main limitations	6
5	Tables and figures	7
5.1	Figure 1 and Table 1: WSB growth and sample construction	7
5.2	Table 2: Determinants of WSB coverage	7
5.3	Table 3, Table 4, and Figure 2: WSB reports and future returns	8
5.4	Table 5 and Figure 3: Pandemic timing versus GME timing	10
5.5	Table 6 and Table 7: Information processing, production, and cash-flow news	11
5.6	Table 8 and Figure 4: Changes in contributor strategies	12
5.7	Table 9, Figure 5, and Table 10: Report types and return predictability	13
5.8	Table 11, Figure 6, and Table 12: Retail trading and investor welfare concerns	15
6	Short summary	17
7	Conclusion	18
7.1	Core conclusion	18
7.2	Economic intuition	18
7.3	Takeaway	18

1 Big picture

- **Question.** Does investment research posted on Reddit’s Wallstreetbets (WSB) contain value-relevant information, and did this value change after the GameStop short squeeze?
- **Object of study.** The paper focuses on single-firm WSB “Due Diligence” (DD) reports, which moderators label as containing analysis and a clear buy or sell signal.
- **Core comparison.** The authors compare DD reports to WSB nonresearch posts and to Seeking Alpha research reports. This distinguishes WSB research quality from general social-media attention.
- **Main pre-GME result.** Before the GameStop event, WSB DD recommendations predict future returns and cash-flow news. In Table 3, one additional net buy DD report predicts a 1.11% one-week return and a 5.17% one-month return in the full sample.
- **Not just GME/AMC.** Excluding GME and AMC, the pre-GME return predictability remains economically meaningful: 0.91% over one week and 2.33% over one month.
- **Post-GME deterioration.** After GameStop, the predictive ability of DD reports largely disappears. The post-GME interaction is negative in one-month return regressions.
- **Mechanism.** The deterioration is linked to a shift in WSB culture. Post-GME reports are more likely to emphasize price pressure, short squeezes, high-attention stocks, and GME/AMC-style narratives rather than firm fundamentals.
- **Cash-flow evidence.** Pre-GME DD reports also predict media sentiment and analyst forecast news, suggesting that the return predictability is not only short-lived price pressure.
- **Retail trading evidence.** DD reports are strongly associated with small retail investor order imbalances. Before GME, small retail imbalances following DD reports predict future returns; after GME, this informativeness disappears or turns negative.
- **Economic takeaway.** WSB initially produced investment research with real information content, but the GameStop event changed contributor incentives and audience composition, reducing research quality and potentially harming less sophisticated retail investors.

2 Data and measurement

2.1 WSB DD reports

- The authors collect Reddit WSB posts from July 2018 through June 2021.
- The main sample is made of DD reports focused on one common stock and containing a directional recommendation.
- Recommendations are manually coded because WSB tickers and recommendations are not standardized.
- If a report is issued outside market hours, its event date is moved to the next trading day.
- The day-0 return is excluded from future-return windows to reduce contamination from simultaneous news and immediate price moves.

Interpretation. The key data-cleaning challenge is that Reddit posts are informal. Manual ticker and recommendation validation is essential because automated parsing can misclassify tickers, sarcasm, or deliberate ticker obfuscation.

2.2 Pre-GME and post-GME periods

- The **pre-GME period** runs from July 2018 through January 12, 2021.
- The **post-GME period** runs from January 14, 2021 through June 2021.
- The paper treats the GameStop short squeeze as a shock to WSB’s size, culture, and contributor composition.
- Figure 1 shows that WSB users increased dramatically around the GameStop episode.

Interpretation. The design is not a randomized experiment. It is a before-after comparison around a salient event that changed WSB participation and attention.

2.3 Main social-media variables

For firm i on day t :

- NetDD_{it} is the number of buy DD reports minus the number of sell DD reports.
- Nonresearch_{it} is the number of WSB nonresearch posts.
- NetSA_{it} is the number of buy Seeking Alpha reports minus sell Seeking Alpha reports.
- PostGME_t equals one in the post-GME period.

Interpretation. NetDD captures directional WSB research. Nonresearch captures WSB attention without research content. NetSA provides an outside benchmark for investment research with more editorial structure.

2.4 Post characteristics: price pressure and attention

The paper classifies DD reports into several types:

- **Price pressure reports:** reports emphasizing short squeezes, coordinated buying, or price-pressure language.
- **Attention reports:** reports about high-attention stocks, measured using high prior absolute returns or high prior WSB posting volume.
- **GME/AMC reports:** reports focusing on the two most salient meme stocks.

Interpretation. These classifications test whether the post-GME decline in informativeness comes from a shift toward speculative, attention-grabbing, or coordination-based content.

2.5 Return, cash-flow-news, and trading outcomes

- Future returns are measured over one week $[1, 5]$ and one month $[1, 21]$ after the report date.
- Longer horizon plots examine return predictability up to 12 weeks.
- Cash-flow news proxies include media sentiment, positive analyst forecast error, and positive forecast revisions.
- Retail trading outcomes are based on small and large retail order imbalances, and institutional imbalances.

Interpretation. The paper combines return predictability, cash-flow news, and order-flow evidence to distinguish information from pure demand pressure.

3 Models and empirical specifications

3.1 Determinants of WSB coverage

The coverage regressions relate WSB and Seeking Alpha activity to firm characteristics:

$$Coverage_{i,t} = \alpha + \beta X_{i,t} + \delta(X_{i,t} \times PostGME_t) + \varepsilon_{i,t}. \quad (1)$$

- *Coverage* can be WSB DD coverage, WSB nonresearch coverage, or Seeking Alpha coverage.
- $X_{i,t}$ includes institutional ownership, breadth of ownership, size, book-to-market, volatility, turnover, returns, media coverage, analyst coverage, high max return, heavy short interest, recent IPO, and GME/AMC indicators.

Interpretation. This establishes what kinds of firms receive WSB attention and whether that changed after GameStop.

3.2 Future-return regression

The main return specification is

$$R_{i,t+1,t+x} = \beta_1 NetDD_{it} + \beta_2 NetDD_{it} \times PostGME_t + \beta_3 Nonresearch_{it} \quad (2)$$

$$+ \beta_4 Nonresearch_{it} \times PostGME_t + \beta_5 NetSA_{it} + \beta_6 NetSA_{it} \times PostGME_t \quad (3)$$

$$+ Controls_{it} + Day_t + \varepsilon_{it}. \quad (2)$$

- $x = 5$ for one-week returns and $x = 21$ for one-month returns.
- β_1 measures DD report informativeness in the pre-GME period.
- β_2 measures the post-GME change in DD informativeness.
- Day fixed effects absorb market-wide news and return movements.
- Standard errors are clustered by firm and month.

Interpretation. The central prediction is $\beta_1 > 0$ if DD reports contain value-relevant information before GME, and $\beta_2 < 0$ if informativeness declines after GME.

3.3 Cash-flow-news regression

The cash-flow regressions use the same structure as Equation (2), but replace returns with cash-flow-news proxies:

$$CFNews_{i,t+1,t+x} = \beta_1 NetDD_{it} + \beta_2 NetDD_{it} \times PostGME_t + \dots + Controls_{it} + Day_t + \varepsilon_{it}. \quad (4)$$

Interpretation. If DD reports predict future cash-flow news, then return predictability is more likely to reflect information production or processing rather than pure temporary price pressure.

3.4 Report-type regressions

The paper partitions DD reports into information-processing versus information-production reports:

$$NetDD_{it} = NetDD_{it}^{Processing} + NetDD_{it}^{Production}. \quad (5)$$

It also partitions DD reports into price-pressure and attention-based reports:

$$NetDD_{it} = NetDD_{it}^{PP/Attention} + NetDD_{it}^{Other}. \quad (6)$$

Interpretation. The first partition asks why pre-GME reports were informative. The second asks why post-GME reports became less informative.

3.5 Retail trading regressions

The main retail-trading specification is

$$R_{i,t+1,t+21} = \beta_1 \text{Imb}_{it} + \beta_2 \text{Imb}_{it} \times \text{PostGME}_t + \beta_3 \text{Imb}_{it} \times \text{DD}_{it} \quad (7)$$

$$+ \beta_4 \text{Imb}_{it} \times \text{DD}_{it} \times \text{PostGME}_t + \beta_5 \text{Imb}_{it} \times \text{NRIndicator}_{it} \quad (8)$$

$$+ \beta_6 \text{Imb}_{it} \times \text{NRIndicator}_{it} \times \text{PostGME}_t + \beta_7 \text{Imb}_{it} \times \text{SA}_{it} \quad (9)$$

$$+ \beta_8 \text{Imb}_{it} \times \text{SA}_{it} \times \text{PostGME}_t + \text{Controls}_{it} + \text{Day}_t + \varepsilon_{it}. \quad (4)$$

Interpretation. β_3 asks whether retail order flow is more informative following DD reports before GME. β_4 asks whether that informativeness changes after GME.

4 Identification and empirical design

4.1 What the paper identifies

- The paper identifies whether WSB DD reports predict future returns, cash-flow news, and retail order imbalances, conditional on rich controls and day fixed effects.
- It does not claim that DD reports are randomly assigned to firms.
- The post-GME analysis identifies a structural break in report informativeness and composition around a major shock to the WSB platform.

Interpretation. The paper’s main evidence is predictive and comparative: WSB DD reports are informative before GME; the same type of posts become less informative after WSB’s culture changes.

4.2 Controls and benchmarks

- Controls include size, book-to-market, prior returns, and prior media sentiment.
- Day fixed effects remove market-wide return variation.
- Nonresearch posts control for general WSB attention.
- Seeking Alpha reports control for general social-finance research outside WSB.
- Many specifications exclude GME and AMC to ensure meme-stock extremes do not drive results.

Interpretation. The comparison to both nonresearch WSB posts and Seeking Alpha is important: the paper is not simply asking whether social media moves stocks, but whether WSB research contains investment value.

4.3 Mechanism tests

- Cash-flow-news regressions test whether DD reports predict fundamentals-related news.
- Information-processing versus production tests distinguish reports built around public information events from reports without obvious news events.
- Price-pressure and attention classifications test whether speculative post-GME strategies explain the decline in informativeness.

- Retail-order-flow tests ask whether DD reports shape small retail trading and whether that trading remains informative.

Interpretation. The evidence points to two forces: DD reports had information content before GME, but post-GME reports increasingly emphasized speculative themes that were less informative.

4.4 Main limitations

- The post-GME event is not randomized.
- The pre-GME period is short relative to more established research platforms.
- DD report recommendations are manually coded, but latent tone and sophistication are hard to measure fully.
- Retail trading data measure order imbalance, not account-level investor welfare.
- Return predictability may reflect both information and delayed price pressure, so the paper relies on multiple tests to separate channels.

Interpretation. The paper is strongest as a triangulated empirical study: return, cash-flow, report-composition, and order-flow evidence all point toward a post-GME deterioration in WSB research quality.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* format. Adjacent figures and tables are grouped to save space and improve comparison. All visuals are directly cropped from the original paper.

5.1 Figure 1 and Table 1: WSB growth and sample construction

Read:

- Figure 1 shows explosive WSB user growth around the GameStop episode.
- Table 1 reports 5,015 DD reports, 3,782 firm-days, and 906 firms in the full DD sample.
- DD reports are mostly buy recommendations: 88% in the full sample, rising from 81% pre-GME to 95% post-GME.
- The share of reports on GME or AMC jumps from 4.18% pre-GME to 19.60% post-GME.
- WSB nonresearch posts are even more GME/AMC-focused after GME, reaching 70.01%.

Economic message. The GameStop event changed both WSB’s size and content. This motivates treating the post-GME period as a distinct regime.

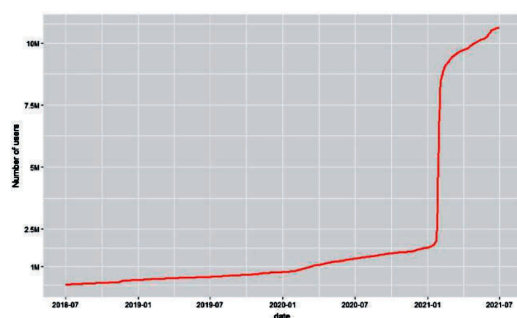


Figure 1
Growth in Reddit's Wallstreetbets

(a) Figure 1: Growth in Reddit's Wallstreetbets

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riptive statistics
SB DD Reports: Full sample

	DD report statistics						Firm-month coverage	
	DD reports	Firm-days	Firms	% buys	% GME or AMC	Posts per contributor	Mean	SD
sample	5,015	3,782	906	88	12.08	1.32	0.04	0.80
GME	2,443	2,096	635	81	4.18	1.33	0.02	0.31
GME	2,572	1,686	501	95	19.60	1.24	0.08	1.57

SB nonresearch reports

	Nonresearch report statistics						Firm-month coverage	
	Nonresearch reports	Firm-days	Firms	% buys	% GME or AMC	Posts per contributor	Mean	SD
sample	13,255	4,656	710	N/A	42.31	1.21	0.10	8.37
GME	5,585	3,344	546	N/A	4.33	1.34	0.05	1.49
GME	7,589	1,312	355	N/A	70.01	1.10	0.27	17.15

eking Alpha reports

	SA Report statistics						Firm-month coverage	
	SA reports	Firm-days	Firms	% buys	% GME or AMC	Posts per contributor	Mean	SD
sample	23,659	22,460	2,955	85	0.48	10.98	0.17	0.71
GME	19,902	18,846	2,652	85	0.39	10.26	0.19	0.76
GME	3,757	3,614	1,630	88	0.96	5.10	0.13	0.30

table reports summary statistics on the sample of social media posts. Panel A reports the results for Due Diligence (DD) reports on Reddit's Wallstreetbets (WSB). DD reports are reports filed by the poster (and verified by the moderator) as containing some analysis and offering a clear buy or sell signal. We limit the sample to DD reports that focus on a single common stock. We report the number of DD reports for the full sample (July 2018 to June 2021), the *Pre-GME* period (July 2018 to January 12, 2021), and the *Post-GME* period (January 14, 2021, to June 2021). For each period, we report the number of firm-days and firms with at least one DD report, the percentage of reports recommending a long position (% buys), the percentage of reports that are written on GME or AMC (GME/AMC), and the average number of DD reports issued by each username (*Posts-per-contributor*). We also report the mean and standard deviation of DD reports across all firm-months. The sample includes 90,175 firm-months in the pre-GME period and 27,344 firm-months in the post-GME period. Panel B reports results for 'NONRESEARCH' posts defined as a post belonging to one of the following WSB categories: *News*, *Losses*, *Gains*, *Charts*, and *Shitpost*, and panel C reports analogous results for the set of research reports provided by Seeking Alpha.

(b) Table 1: Descriptive statistics

5.2 Table 2: Determinants of WSB coverage

Read:

- WSB DD coverage is negatively related to institutional ownership, consistent with WSB targeting less institutionally held stocks.
- WSB coverage is related to small, volatile, attention-grabbing, and meme-related stocks.
- Seeking Alpha coverage is more strongly related to traditional coverage variables such as size and analyst/media coverage.
- Post-GME interactions show that WSB coverage changed materially after GameStop.

Economic message. WSB research is not just another version of traditional analyst-style coverage. It focuses on a different set of stocks, and the focus shifts after the GameStop event.

Table 2
Determinants of WSB coverage

	WSB DD coverage [1]	WSB Nonresearch coverage [2]	SA coverage [3]
<i>Inst ownership</i>	-0.03 (-3.84)	-0.04 (-4.57)	-0.06 (-5.87)
<i>Inst ownership</i> $\times$ <i>Post-GME</i>	-0.04 (-1.55)	0.01 (0.72)	0.04 (3.94)
<i>log(Breadth of own.)</i>	0.01 (0.80)	0.01 (0.31)	0.03 (2.12)
<i>log(Breadth of own.)</i> $\times$ <i>Post-GME</i>	-0.01 (-0.58)	-0.01 (-0.83)	-0.02 (-1.46)
<i>log(Size)</i>	0.26 (4.56)	0.38 (4.26)	0.84 (12.28)
<i>log(Size)</i> $\times$ <i>Post-GME</i>	0.50 (3.79)	-0.03 (-0.23)	-0.10 (-1.36)
<i>log(BM)</i>	-0.02 (-2.39)	-0.03 (-2.05)	-0.03 (-2.42)
<i>log(BM)</i> $\times$ <i>Post-GME</i>	0.03 (1.96)	0.02 (1.97)	-0.02 (-1.71)
<i>Negative BM</i>	0.02 (1.21)	0.03 (0.95)	0.02 (0.53)
<i>Negative BM</i> $\times$ <i>Post-GME</i>	-0.03 (-0.27)	0.00 (-0.03)	0.01 (0.28)
<i>log(Vol)</i>	0.11 (4.74)	0.15 (3.71)	0.47 (9.51)
<i>log(Vol)</i> $\times$ <i>Post-GME</i>	0.16 (2.42)	-0.03 (-0.44)	-0.26 (-4.46)
<i>log(Turn)</i>	0.00 (0.20)	0.01 (0.53)	-0.03 (-1.98)
<i>log(Turn)</i> $\times$ <i>Post-GME</i>	0.12 (3.01)	0.05 (1.84)	0.08 (5.23)
<i>Return [m-1]</i>	0.01 (2.08)	0.02 (3.98)	0.01 (2.12)
<i>Return [m-1]</i> $\times$ <i>Post-GME</i>	0.04 (1.50)	0.04 (1.66)	0.01 (1.01)
<i>Return [m-2, m-12]</i>	0.30 (3.01)	0.33 (2.43)	0.23 (4.95)
<i>Return [m-2, m-12]</i> $\times$ <i>Post-GME</i>	-0.27 (-2.85)	-0.31 (-2.36)	-0.21 (-4.79)
<i>log(Media coverage)</i>	0.17 (3.68)	0.25 (3.60)	0.38 (6.24)
<i>log(Media coverage)</i> $\times$ <i>Post-GME</i>	0.01 (0.12)	-0.19 (-3.41)	-0.18 (-3.45)
<i>log(IBES coverage)</i>	-0.01 (-0.76)	-0.02 (-1.12)	0.04 (2.87)
<i>log(IBES coverage)</i> $\times$ <i>Post-GME</i>	-0.08 (-2.59)	-0.02 (-0.74)	-0.05 (-3.02)
<i>High max</i>	0.03 (2.87)	0.05 (3.35)	0.06 (3.62)
<i>High max</i> $\times$ <i>Post-GME</i>	0.14 (6.36)	-0.02 (-0.56)	0.01 (0.66)
<i>Heavy short</i>	0.05 (2.69)	0.06 (2.19)	0.15 (5.53)
<i>Heavy short</i> $\times$ <i>Post-GME</i>	0.27 (3.46)	0.07 (0.71)	-0.06 (-1.64)
<i>Recent IPO</i>	0.57 (3.80)	0.75 (4.15)	0.97 (6.92)
<i>Recent IPO</i> $\times$ <i>Post-GME</i>	1.17 (3.49)	0.01 (0.03)	0.41 (2.92)
<i>GME/AMC</i>	3.38 (8.52)	2.76 (2.96)	1.77 (4.14)

(a) Table 2, first page

Table 2
(Continued)

	WSB DD coverage [1]	WSB Nonresearch coverage [2]	SA coverage [3]
<i>GME/AMC</i> $\times$ <i>Post-GME</i>	15.38 (12.43)	21.71 (18.64)	0.76 (1.53)
Obs. (firm-months)	113,728	113,728	113,728
Fixed effects	Month	Month	Month
R-squared (%)	3.28	3.94	13.34

This table presents the estimates from Equation (1):

$$Coverage_{it} = \alpha + \beta_1 Chars_{it-1} + \beta_2 Chars_{it-1} \times Post_t + Month_t + \epsilon_{it}.$$

The dependent variable, *Coverage*, is either *WSB DD Coverage* defined as $\log(1 + \text{DD reports})$ for firm i during month t , *WSB Nonresearch coverage*, defined as $\log(1 + \text{Nonresearch posts})$ for firm i during month t , or *SA coverage* defined as $\log(1 + \text{SA reports})$ for firm i during month t . *Chars* include the following firm characteristics: the percentage of the firm's shares held by institutional investors at the end of the prior year (*Inst. ownership*), the number of common shareholders (*Breadth of ownership*), market capitalization (*Size*), book-to-market ratio (*BM*), return volatility (*Volatility*), share turnover (*Turnover*), returns over the prior month (*Ret_{m-1}*), returns over the prior 2 to 12 months (*Ret_{m-2,m-12}*), the number of media articles mentioning the firm in the prior year (*Media coverage*), the number of sell-side analysts issuing a forecast for the firm in the prior year (*IBES coverage*), an indicator equal to one if the maximum daily return of the firm in the prior month is in the top quintile (*High max*), an indicator equal to one if the firm is in the top quintile of short interest (*Heavy short*), an indicator equal to one if the firm went public in the past 6 months (*Recent IPO*), and an indicator equal to one for GME or AMC (*GME/AMC*). We allow the loadings on firm characteristics to vary in the pre-GME and post-GME periods by interacting the firm characteristics with *Post*, an indicator equal to one for the Post-GME period (February 2021 to June of 2021) and zero for the Pre-GME period (July 2018 to December 2020), with the month of the GME-event (January 2021) excluded. We also include calendar-month fixed effects, and we standardize all continuous variables to have mean zero and unit variance. Appendix C defines the variables in detail. Standard errors are clustered by firm and month, and t -statistics are reported in parentheses.

In Specification 3, we examine the determinants of SA coverage. Perhaps surprisingly, in the pre-GME period, we find that *SA* and *WSB* have similar, strong preferences for lottery-like stocks, stocks with heavy short-interest and stocks that recently went public. However, in sharp contrast to *WSB DD*

(b) Table 2, continued

5.3 Table 3, Table 4, and Figure 2: WSB reports and future returns

Read:

- Table 3 shows that pre-GME DD reports predict future returns: 1.11% over one week and 5.17% over one month in the full sample.
- Excluding GME and AMC, DD reports still predict 0.91% one-week returns and 2.33% one-month returns.
- The interaction $\text{NetDD} \times \text{Post}$ is negative for one-month returns, indicating a sharp deterioration after GME.
- Table 4 shows that pre-GME return predictability persists over longer horizons rather than immediately reversing.
- Figure 2 visualizes this persistence over horizons up to 12 weeks.

Economic message. The pre-GME evidence is consistent with information in DD reports, not simply short-run price pressure. The post-GME evidence shows that this informativeness largely disappears.

Table 3
WSB reports and future returns

	Ret [1,5] [1]	Ret [1,21] [2]	Ret [1,5] [3]	Ret [1,21] [4]
<i>Net DD</i>	1.11% (-2.17)	5.17% (-2.77)	0.91% (2.12)	2.33% (-2.46)
<i>Net DD × Post-GME</i>	0.11% (0.09)	-5.30% (-3.27)	-0.51% (-0.83)	-3.45% (-2.52)
<i>WSB Nonresearch</i>	0.20% (1.44)	3.61% (1.33)	0.14% (1.08)	0.40% (1.03)
<i>WSB Nonresearch × Post-GME</i>	1.06% (7.05)	-2.84% (-1.25)	-0.44% (-2.89)	-1.68% (-3.80)
<i>Net SA</i>	0.33% (4.24)	0.59% (2.48)	0.32% (4.26)	0.65% (2.68)
<i>Net SA × Post-GME</i>	0.11% (0.69)	0.39% (0.95)	-0.01% (-0.09)	0.36% (0.91)
<i>log(Size)</i>	-0.08% (-1.61)	-0.27% (-1.27)	-0.08% (-1.61)	-0.27% (-1.26)
<i>log(BM)</i>	-0.07% (-0.90)	-0.24% (-0.80)	-0.07% (-0.92)	-0.25% (-0.83)
<i>Ret [0]</i>	-7.32% (-5.19)	-9.54% (-4.70)	-7.32% (-5.20)	-9.44% (-4.58)
<i>Ret [-5, -1]</i>	-2.82% (-2.66)	-4.11% (-2.78)	-2.77% (-2.58)	-3.99% (-2.70)
<i>Ret [-26, -6]</i>	-0.34% (-1.18)	-0.56% (-0.47)	-0.32% (-1.06)	-0.70% (-0.56)
<i>News sentiment [0]</i>	0.10% (3.25)	0.10% (1.09)	0.07% (2.12)	0.08% (0.97)
<i>News sentiment [-5, -1]</i>	0.02% (0.56)	0.03% (0.34)	0.01% (0.23)	0.02% (0.30)
<i>News sentiment [-26, -6]</i>	0.01% (0.38)	0.05% (0.79)	0.01% (0.64)	0.06% (1.04)
<i>Net DD + Net DD × Post</i>	1.22% (0.96)	-0.13% (-0.11)	0.40% (0.81)	-1.12% (-1.05)
<i>Net DD - Nonresearch</i>	0.91% (1.84)	1.56% (0.87)	0.77% (1.77)	1.93% (2.24)
<i>Net DD × Post - Nonresearch × Post</i>	-0.95% (-1.63)	-2.45% (-1.19)	-0.07% (-0.12)	-1.78% (-1.34)
<i>Net DD - Net SA</i>	0.79% (1.55)	4.58% (2.47)	0.59% (1.35)	1.68% (1.72)
<i>Net DD × Post - Net SA × Post</i>	0.00% (0.00)	-5.68% (-3.11)	-0.50% (-0.77)	-3.81% (-3.07)
Obs. (firm-days)	2,772,053	2,772,053	2,770,545	2,770,545
Day FE	Yes	Yes	Yes	Yes
Include GME & AMC	Yes	Yes	No	No

This table reports results from the estimation of Equation (2):

$$\begin{aligned}
R_{it+1,t+x} = & \beta_1 \text{Net DD}_{it} + \beta_2 \text{Net DD}_{it} \times \text{Post}_t + \beta_3 \text{Nonresearch}_{it} \\
& + \beta_4 \text{Nonresearch}_{it} \times \text{Post}_t + \beta_5 \text{Net SA}_{it} + \beta_6 \text{Net SA}_{it} \times \text{Post}_t \\
& + \text{Control}_{sit} + \text{Day}_t + \varepsilon_{it}.
\end{aligned}$$

The dependent variable, R_{it} , is the stock return measured over the subsequent week (i.e., $x=5$ trading days) or the subsequent month ($x=21$ trading days). *Net DD* is the number of buy DD recommendations less the number of sell DD recommendations for stock i on day t , and *Net DD × Post*, interacts *Net DD* with an indicator for the *Post-GME* period as defined in Table 1. *Nonresearch* is the number of nonresearch posts (as defined in Table 1) and *Net SA* is the number of SA reports issuing a buy recommendation less the number of SA reports issuing a sell recommendation for stock i on day t . *Controls* include size, book-to-market, prior returns and prior media

(a) Table 3: Future returns

Table 4
WSB Reports and Future Returns: Alternative Horizons

Panel A: Include GME & AMC

Ret. Period	Net DD		Net DD × Post GME		Net DD + Net DD × Post GME	
	Coefficient	t-stat	Coefficient	t-stat	Coefficient	t-stat
0	0.88%	(6.75)	-0.57%	(-3.19)	0.31%	(2.52)
1	0.23%	(1.61)	0.32%	(1.13)	0.55%	(2.36)
2	0.33%	(1.89)	-0.20%	(-0.84)	0.13%	(0.72)
3	0.18%	(1.80)	-0.29%	(-1.62)	-0.11%	(-0.72)
4	0.10%	(0.87)	0.36%	(0.76)	0.45%	(0.97)
5	0.20%	(2.32)	-0.14%	(-1.36)	0.05%	(0.57)
[6-10]	0.90%	(1.33)	-0.39%	(-0.76)	0.51%	(0.72)
[11-15]	1.27%	(2.08)	-1.53%	(-2.01)	-0.26%	(-0.72)
[16-20]	1.61%	(1.46)	-2.11%	(-1.73)	-0.50%	(-0.88)
[21-60]	4.41%	(3.11)	-2.46%	(-1.01)	1.95%	(0.94)

Panel B: Exclude GME & AMC

Ret. Period	Net DD		Net DD × Post GME		Net DD + Net DD × Post GME	
	Coefficient	t-stat	Coefficient	t-stat	Coefficient	t-stat
0	0.89%	(6.81)	-0.40%	(-2.36)	0.49%	(4.45)
1	0.24%	(1.53)	0.24%	(1.21)	0.48%	(4.33)
2	0.25%	(1.94)	-0.13%	(-0.71)	0.12%	(0.89)
3	0.15%	(1.57)	-0.21%	(-1.32)	-0.06%	(-0.51)
4	0.11%	(0.97)	0.03%	(0.12)	0.14%	(0.76)
5	0.18%	(2.73)	-0.22%	(-1.82)	-0.05%	(-0.44)
[6-10]	0.29%	(1.28)	-0.28%	(-0.75)	0.01%	(0.03)
[11-15]	0.49%	(1.76)	-0.91%	(-2.34)	-0.42%	(-1.83)
[16-20]	0.60%	(1.32)	-1.42%	(-1.86)	-0.82%	(-1.29)
[21-60]	-0.44%	(-0.35)	-0.32%	(-0.18)	-0.76%	(-0.54)

This table repeats the analysis in Table 3 after replacing the dependent variable with the return on the current day (i.e., Day 0), each of the subsequent five days, the subsequent 2, 3, and 4 weeks, and the subsequent 5 through 12 weeks. We report the estimate on *Net DD*, *Net DD × Post*, and the sum of the two estimates. Panel A reports the results for the full sample, and Panel B reports analogous results after excluding GME and AMC. Standard errors are clustered by firm and month, and t -statistics are reported next to each estimate.

(b) Table 4: Alternative horizons

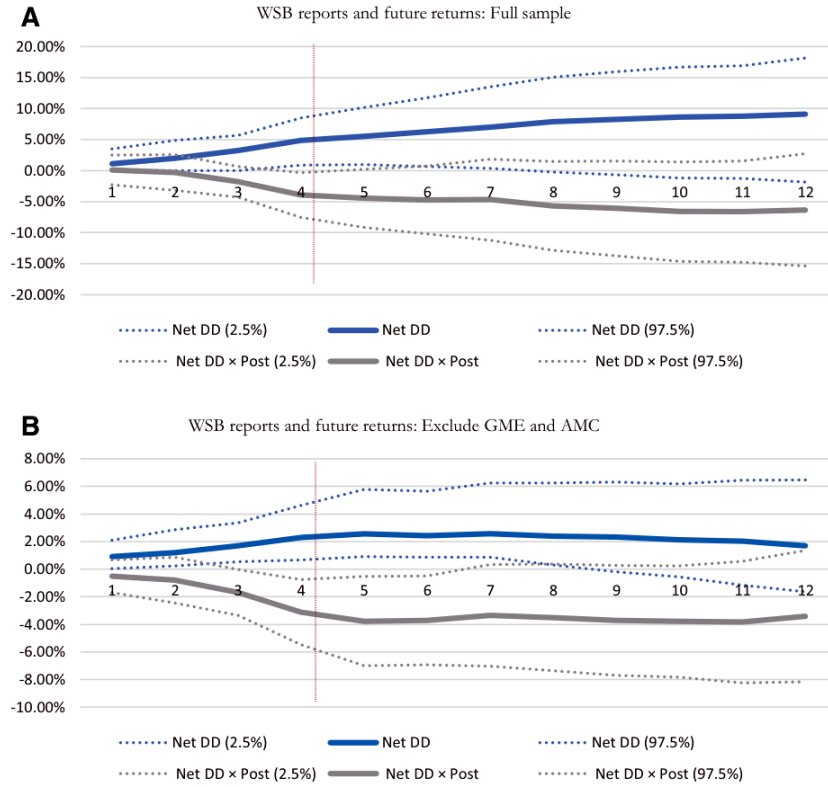


Figure 2

Figure 4: Figure 2: WSB reports and future returns by week

5.4 Table 5 and Figure 3: Pandemic timing versus GME timing

Read:

- Table 5 separates prepandemic, pandemic/pre-GME, and post-GME effects.
- DD report informativeness is positive both before and during the pandemic but before GameStop.
- Figure 3 shows positive return predictability across 2020 quarters and a sharp decline in 2021.
- The decline is therefore more closely linked to the GME regime shift than to the pandemic itself.

Economic message. The deterioration in DD informativeness is not simply a COVID-market phenomenon. It aligns with the GameStop-driven change in WSB culture.

Table 5
WSB reports and future returns: The role of the pandemic

	Ret [1,5] [1]	Ret [1,21] [2]	Ret [1,5] [3]	Ret [1,21] [4]
<i>Net DD</i> $\times$ <i>Prepandemic</i>	1.12% (2.18)	3.72% (2.02)	1.13% (2.18)	3.75% (1.99)
<i>Net DD</i> $\times$ <i>Post-pandemic</i>	1.09% (1.58)	5.01% (2.70)	0.80% (1.45)	1.67% (1.79)
<i>Net DD</i> $\times$ <i>Post-GME</i>	0.13% (0.11)	-5.14% (-3.23)	-0.34% (-0.56)	-2.79% (-2.07)
<i>WSB Nonresearch</i> $\times$ <i>Prepandemic</i>	0.14% (0.87)	0.67% (1.08)	0.13% (0.83)	0.66% (1.06)
<i>WSB Nonresearch</i> $\times$ <i>Post-pandemic</i>	0.25% (1.40)	6.03% (1.36)	0.15% (0.92)	0.13% (0.44)
<i>WSB Nonresearch</i> $\times$ <i>Post-GME</i>	1.01% (3.46)	-5.26% (-1.38)	-0.45% (-2.40)	-1.41% (-3.98)
<i>Net SA</i> $\times$ <i>Prepandemic</i>	0.33% (3.42)	0.73% (2.57)	0.33% (3.44)	0.74% (2.59)
<i>Net SA</i> $\times$ <i>Post-pandemic</i>	0.32% (3.49)	0.44% (1.23)	0.30% (3.90)	0.49% (1.39)
<i>Net SA</i> $\times$ <i>Post-GME</i>	0.12% (0.76)	0.53% (1.06)	0.08% (0.05)	0.52% (1.06)
<i>Net DD (Prepandemic - Post-pandemic)</i>	0.02% (0.03)	-0.49% (-0.67)	0.33% (0.46)	2.08% (0.98)
<i>WSB Nonresearch (Prepandemic - Post-pandemic)</i>	-0.12% (-0.67)	-5.35% (-1.18)	-0.03% (-0.14)	0.53% (1.03)
<i>Net SA (Prepandemic - Post-pandemic)</i>	0.02% (0.17)	0.29% (0.74)	0.03% (0.33)	0.25% (0.67)
Obs. (firm-days)	2,772,053	2,772,053	2,770,545	2,770,545
Day FE	Yes	Yes	Yes	Yes
Table 3 controls	Yes	Yes	Yes	Yes
Include GME & AMC	Yes	Yes	No	No

This table repeats Table 3 after partitioning *Net DD* into *Net DD* $\times$ *Prepandemic* and *Net DD* $\times$ *Post-pandemic* where *Prepandemic* is an indicator equal to one for July 2018 to February 25, 2020, and zero otherwise, an *Post-pandemic* is an indicator equal to one for the pre-GME period that coincides with the pandemic (February 26, 2020, to January 12, 2021) and the post-GME period. We also create analogous *Prepandemic* and *Post-pandemic* variables for *WSB Nonresearch* and *Net SA*. The regression includes all the controls from Table 3, but in the interest of brevity, the estimates on controls are unreported. We also include calendar-day fixed effect and we standardize all continuous variables to have mean zero and unit variance. Standard errors are clustered by firm and month, and *t*-statistics are reported in parentheses. Below the regression estimates, we report formal tests for whether certain pairs of coefficients are significantly different from zero.

(a) Table 5: Role of the pandemic

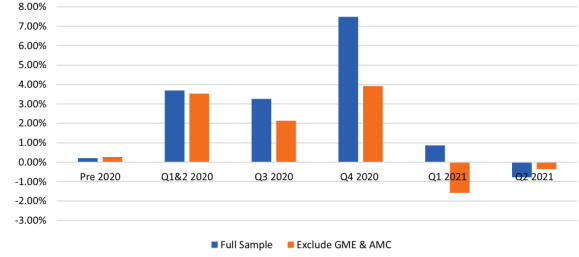


Figure 3
WSB reports and future returns: Results by calendar quarter

This figure reports the estimates from Specifications 2 and 4 of Table 3 over the following intervals: the pre-2020 sample, quarters 1 and 2 of 2020, and each of the remaining quarters in 2020 and 2021. We include the small number of pre-GME DD reports in January of 2021 in the Q4 2020 estimates. The blue bars represent the estimates for the full sample of stocks (i.e., Specification 2 of Table 3), and the orange bars represent the estimates for the sample that excludes GME and AMC (i.e., Specification 4 of Table 3).

the informativeness of *WSB* reports in the pre-GME period would generalize to other market environments.¹⁸

3.2 Information processing versus information production

Next, we explore the economic channel underlying the investment value of

(b) Figure 3: Results by calendar quarter

5.5 Table 6 and Table 7: Information processing, production, and cash-flow news

Read:

- Table 6 shows that both information-processing and information-production DD reports predict returns before GME.
- Net DD production predicts 5.99% one-month returns in the full sample before GME, while processing predicts 3.95%.
- Post-GME interactions are negative, especially for one-month production reports.
- Table 7 shows that DD reports predict cash-flow news proxies before GME, including media sentiment and analyst forecast news.
- The cash-flow-news relation weakens or turns negative after GME.

Economic message. Pre-GME *WSB* DD reports appear to contain real value-relevant information. Post-GME, the research-production process becomes much less informative.

Table 6

WSB reports and future returns: Information processing versus information production

	Ret [1,5] [1]	Ret [1,21] [2]	Ret [1,5] [3]	Ret [1,21] [4]
<i>Net DD Processing</i>	0.69% (0.84)	3.95% (2.09)	0.32% (0.50)	2.27% (2.05)
<i>Net DD Processing × Post-GME</i>	-0.09% (-0.06)	-2.00% (-0.88)	0.07% (0.08)	-2.52% (-1.69)
<i>Net DD Production</i>	1.41% (2.93)	5.99% (2.99)	1.35% (2.79)	2.39% (2.52)
<i>Net DD Production × Post-GME</i>	0.09% (0.06)	-7.03% (-5.39)	-0.94% (-1.37)	-3.87% (-2.66)
<i>WSB nonresearch</i>	0.19% (1.38)	3.22% (1.32)	0.14% (1.02)	0.42% (1.03)
<i>WSB nonresearch × Post-GME</i>	1.11% (5.38)	-2.56% (-1.26)	-0.43% (-2.74)	-1.70% (-3.67)
<i>Net SA</i>	0.31% (4.05)	0.56% (2.33)	0.31% (4.10)	0.63% (2.54)
<i>Net SA × Post-GME</i>	0.12% (0.75)	0.44% (1.09)	-0.01% (-0.03)	0.39% (0.99)
<i>Net DD (processing - production)</i>	-0.72% (-0.90)	-2.03% (1.44)	-1.03% (1.46)	-0.13% (0.14)
<i>Net DD (processing - production) × Post-GME</i>	-0.18% (-0.09)	5.04% (3.80)	1.01% (-1.18)	1.35% (1.06)
Obs. (firm-days)	2,772,053	2,772,053	2,770,545	2,770,545
Day FE	Yes	Yes	Yes	Yes
Table 3 controls	Yes	Yes	Yes	Yes
Include GME & AMC	Yes	Yes	No	No

This table repeats Table 3 after partitioning all DD reports into information processing versus information production reports. *Processing* is an indicator equal to one if the report is issued around a major information event, defined as an earnings announcement issued on the previous or current day (-1, 0), abnormal media coverage on the previous or current day, or an analyst revision on the previous or current day. *Net DD production* is the *Net DD* measure computed for the subset of reports where *Processing* = 1, and *Net DD production* is the *Net DD* measure computed for the subset of reports where *Processing* = 0. All other variables are defined as in Table 3. The regression includes all the controls from Table 3, but in the interest of brevity, the estimates are not reported. We also include calendar-day fixed effects, and we standardize all continuous variables to have mean zero and unit variance. Standard errors are clustered by firm and month, and *t*-statistics are reported in parentheses. Below the regression estimates, we also report a formal test of whether certain pairs of coefficient are significantly different from zero.

processing, with the remaining 60% of reports classified as information production. We then repeat Equation (2) after partitioning *Net DD* into *Net DD processing* and *Net DD production*.

(a) Table 6: Processing versus production

Table 7

WSB reports and future cash-flow news

	Media [1,5] [1]	Media [1,21] [2]	Pos FE [1,5] [3]	Pos FE [1,21] [4]	Pos FR [1,5] [3]	Pos FR [1,21] [4]
<i>Net DD</i>	5.38% (2.37)	18.49% (2.62)	4.22% (2.37)	3.51% (1.62)	2.22% (1.73)	1.92% (1.52)
<i>Net DD × Post-GME</i>	-4.71% (-2.05)	-25.37% (-3.24)	-17.11% (-5.94)	-13.86% (-5.65)	-3.00% (-2.34)	-2.83% (-1.96)
<i>WSB NonResearch</i>	-1.19% (-1.27)	-2.70% (-0.53)	0.34% (0.16)	-2.16% (-1.40)	0.55% (0.95)	0.14% (0.19)
<i>WSB NonResearch × Post-GME</i>	0.10% (0.11)	-1.75% (-0.35)	-0.58% (-0.26)	1.64% (0.79)	-1.06% (-1.76)	-1.17% (-1.74)
<i>Net SA</i>	-0.84% (-1.26)	-7.94% (-2.83)	-1.83% (-1.65)	-0.16% (-0.18)	-0.15% (-0.57)	-0.10% (-0.28)
<i>Net SA × Post-GME</i>	0.62% (0.67)	2.39% (0.82)	1.66% (0.42)	1.11% (0.74)	0.58% (0.75)	0.42% (0.31)
<i>log(Size)</i>	0.65% (5.47)	3.21% (5.66)	3.00% (10.15)	3.52% (11.04)	-0.09% (-0.44)	0.14% (0.49)
<i>log(BM)</i>	-0.85% (-6.62)	-4.26% (-6.75)	-1.37% (-2.05)	-0.90% (-1.13)	-0.53% (-3.37)	-1.17% (-3.80)
<i>Ret [0]</i>	18.90% (7.93)	31.06% (7.73)	15.55% (3.15)	13.40% (6.01)	9.18% (7.67)	13.10% (8.72)
<i>Ret [-5, -1]</i>	2.94% (4.37)	6.21% (2.54)	10.75% (2.35)	11.18% (4.74)	5.94% (8.35)	10.28% (9.76)
<i>Ret [-26, -6]</i>	0.48% (1.37)	3.49% (1.98)	8.19% (3.97)	5.89% (2.21)	4.07% (5.35)	7.27% (6.32)
<i>News sentiment [0]</i>	33.26% (34.13)	87.09% (25.06)	2.17% (3.18)	2.08% (4.01)	2.43% (7.93)	2.49% (9.86)
<i>News sentiment [-5, -1]</i>	15.53% (22.72)	56.24% (20.96)	0.98% (1.85)	1.61% (4.00)	1.03% (10.51)	1.35% (10.23)
<i>News sentiment [-26, -6]</i>	7.90% (18.59)	30.76% (16.15)	0.95% (3.93)	0.75% (3.20)	0.33% (5.49)	0.61% (6.25)
<i>Net DD + Net DD × Post-GME</i>	0.67% (1.33)	-6.88% (-1.83)	-12.89% (-4.94)	-10.35% (-7.67)	-0.78% (-1.50)	-0.91% (-1.21)
<i>Net DD - Nonresearch</i>	6.57% (2.75)	21.19% (2.94)	3.88% (1.27)	5.67% (2.40)	1.67% (1.26)	1.78% (1.25)
<i>Net DD × Post - Nonresearch × Post</i>	-4.81% (-2.01)	-23.62% (-3.03)	-16.53% (-4.64)	-15.50% (-4.63)	-1.94% (-1.50)	-1.66% (-1.07)
<i>Net DD - Net SA</i>	6.22% (2.63)	26.43% (-3.37)	6.05% (3.00)	3.67% (1.56)	2.37% (1.79)	2.02% (1.62)
<i>Net DD × Post - Net SA × Post</i>	-5.33% (-2.18)	-27.76% (-3.05)	-18.77% (-4.34)	-14.97% (-5.62)	-3.58% (-2.11)	-3.26% (-1.62)
Obs. (firm-days)	2,772,053	2,772,053	164,018	642,522	1,964,351	1,964,351
Day FE	Yes	Yes	Yes	Yes	Yes	Yes
Include GME & AMC	Yes	Yes	Yes	Yes	Yes	Yes

This table repeats the analysis in Table 3 after replacing the dependent variable, returns, with one of three proxies for cash-flow news, measured over the subsequent week (i.e., $x = 5$ trading days) or the subsequent month ($x = 2$ trading days). Cash-flow news is measured as either *Media Sentiment (Media)*, computed as the sum of the daily Bloomberg sentiment score; *Positive Forecast Error (Pos FE)*, an indicator equal to one if the realized earnings exceed the median quarterly forecast across all I/B/E/S analysts as of day t ; and *Positive Forecast Revision (Pos FR)* computed as the number of upward revisions by I/B/E/S analysts scaled by the total number of revisions. All other variables are defined as in Table 3. More detailed variable definitions are available in Appendix C. Specifications 1 and 2 report the results for the full sample for 5-day and 21-day measures of *Media Sentiment*. Specifications 3 and 4 (5 and 6) report analogous results for *Positive Forecast Error* (*Positive Forecast Revision*).

(b) Table 7: Future cash-flow news

5.6 Table 8 and Figure 4: Changes in contributor strategies

Read:

- Table 8 shows that price-pressure reports increase sharply post-GME.
- The baseline price-pressure indicator rises by 19.40 percentage points after GME.
- High-attention reports also increase substantially after GME.
- Figure 4 shows the shift over time: price-pressure and high-attention reports rise sharply in Q1 and Q2 of 2021.

Economic message. The post-GME WSB environment becomes more focused on attention and price-pressure narratives, which helps explain the decline in investment value.

Table 8
Changes in contributor strategies in the post-GME period

<i>A. Price pressure reports</i>				
	<i>PP</i> [1]	<i>PP</i> [2]	<i>PP2</i> [3]	<i>PP2</i> [4]
<i>Intercept</i>	11.63% (3.29)	7.91% (6.00)	21.74% (5.35)	17.31% (11.63)
<i>Post-GME</i>	19.40% (7.45)	17.25% (5.67)	28.51% (10.07)	29.67% (9.44)
Obs. (DD reports)	5,015	4,409	5,015	4,409
Include GME & AMC	Yes	No	Yes	No
<i>B. Attention-based reports</i>				
	<i>High attention</i> [1]	<i>High attention</i> [2]	<i>High absolute return</i> [3]	<i>High WSB posts</i> [4]
<i>Intercept</i>	32.53% (10.00)	30.96% (10.10)	21.16% (11.88)	17.47% (5.21)
<i>Post-GME</i>	24.55% (5.70)	20.46% (4.96)	12.25% (3.39)	25.49% (4.85)
Obs. (DD reports)	5,015	4,409	5,015	5,015
Include GME & AMC	Yes	No	Yes	Yes

This table reports estimates from the following regression:

$$Y_{it} = \alpha + \beta_1 Post_{it} + \epsilon_{it}$$

The dependent variable, Y_{it} , is an indicator that classifies a DD report as either a price pressure report (panel A) or an attention-based report (panel B). We classify a report as a price pressure report if either (1) the number of price pressure words exceed the number of fundamental words (*PP*) or (2) the report contains at least one price pressure word (*PP2*). Appendix C provides the list of price pressure and fundamental words. We classify report as attention-based (*High attention*) if either (1) the absolute return on the day prior to the DD report was in the top decile (*High absolute return*) or (2) the firm had more than one nonresearch post issued on WSB over the previous 5 trading days (*High WSB posts*). We also report the results separately for *High absolute return* and *High WSB posts*. The independent variable, *Post*, is an indicator equal to one for the *Post-GME* period, as defined in Table 1, and zero otherwise. Standard errors are clustered by firm and date, and *t*-statistics are reported below each estimate.

(a) Table 8: Contributor strategies

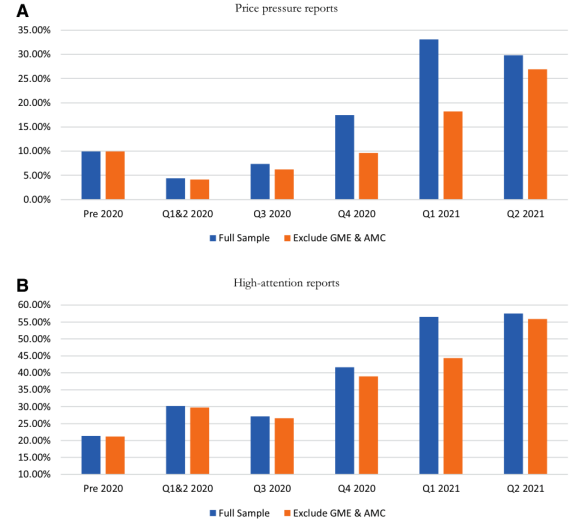


Figure 4
The frequency of price pressure and high attention reports by calendar quarter

(b) Figure 4: Price-pressure and attention reports

5.7 Table 9, Figure 5, and Table 10: Report types and return predictability

Read:

- Table 9 shows that price-pressure and attention reports are informative before GME but much less informative after GME.
- Price-pressure DD reports predict high one-month returns before GME in the full sample, but the post-GME interaction is strongly negative.
- Attention reports also show a negative post-GME interaction.
- Figure 5 shows that the post-GME deterioration persists across horizons.
- Table 10 confirms that price-pressure/attention reports do not show robust positive post-GME predictability over alternative return horizons.

Economic message. The culture shift is central: the types of reports that expanded after GME are the same types whose informativeness deteriorated.

Table 9
WSB reports and future returns: Price pressure and attention reports

	<i>Ret</i> [1,21] [1]	<i>Ret</i> [1,21] [2]	<i>Ret</i> [1,21] [3]	<i>Ret</i> [1,21] [4]	<i>Ret</i> [1,21] [5]	<i>Ret</i> [1,21] [6]
<i>Net DD</i>	2.11% (1.98)	2.17% (2.19)	1.88% (2.35)	1.41% (2.12)	1.41% (2.03)	1.31% (2.10)
<i>Net DD</i> × <i>Post-GME</i>	0.05% (0.03)	−2.53% (−1.46)	−0.61% (−0.36)	−0.81% (−0.54)	−0.15% (−0.14)	−0.45% (−0.35)
<i>Net DD PP</i>	31.73% (2.03)	1.93% (1.33)			8.88% (2.40)	
<i>Net DD PP</i> × <i>Post-GME</i>	−35.80% (−2.15)	−3.24% (−1.85)				
<i>Net DD attention</i>			8.57% (2.25)	2.55% (1.60)		
<i>Net DD attention</i> × <i>Post-GME</i>			−10.58% (−3.10)	−5.16% (−2.84)		
<i>Net DD PP</i> or <i>Attention</i>					8.88% (2.40)	2.57% (1.82)
<i>Net DD PP</i> or <i>Attention</i> × <i>Post</i>					−10.59% (−3.24)	−5.08% (−3.22)
<i>WSB nonresearch</i>	3.44% (1.40)	0.40% (1.03)	3.31% (1.30)	0.33% (0.89)	3.32% (1.29)	0.33% (0.89)
<i>WSB nonresearch</i> × <i>Post-GME</i>	−2.71% (−1.27)	−1.64% (−3.75)	−2.45% (−1.14)	−1.51% (−3.57)	−2.50% (−1.17)	−1.54% (−3.63)
<i>Net SA</i>	0.60% (2.58)	0.65% (2.69)	0.59% (2.52)	0.65% (2.69)	0.60% (2.52)	0.65% (2.70)
<i>Net SA</i> × <i>Post-GME</i>	0.36% (0.88)	0.35% (0.89)	0.38% (0.94)	0.36% (0.91)	0.38% (0.93)	0.35% (0.90)
Obs. (firm-days)	2,772,053	2,770,545	2,772,053	2,770,545	2,772,053	2,770,545
Day FE	Yes	Yes	Yes	Yes	Yes	Yes
Table 3 controls	Yes	Yes	Yes	Yes	Yes	Yes
Include GME & AMC	Yes	No	Yes	No	Yes	No

This table examines the incremental informativeness of price pressure reports and high-attention reports. We classify a report as a price pressure report if the number of price pressure words exceed the number of fundamental words (see Appendix C for the list of price pressure and fundamental words). We classify a report as *High attention* if either (1) the absolute return on the day prior to the DD report was in the top decile (*High absolute return*) or (2) the firm had more than one nonresearch post issued on WSB over the previous 5 trading days (*High WSB posts*); and we classify a report as *PP* or *Attention* if the report is classified as either *Price pressure* or *High attention*. *Net DD PP* is the *Net DD* measure computed for the subset of reports classified as price pressure, and *Net DD attention* and *Net DD PP* or *Attention* are computed analogously. Table 3 defines other variables. The regression includes all the controls from Table 3, but in the interest of brevity, the estimates for controls are unreported. We also include calendar-day fixed effects, and we standardize all continuous variables to have mean zero and unit variance. Standard errors are clustered by firm and month, and *t*-statistics are reported in parentheses.

(a) Table 9: Price pressure and attention reports

Table 10
WSB price pressure and attention reports: Alternative return horizons

<i>Ret. Period</i>	<i>Net DD PP</i> or <i>Attention</i>		<i>Net DD PP</i> or <i>Attention</i> or <i>Post-GME</i>		<i>Net DD PP</i> or <i>Attention</i> + <i>Net DD PP</i> or <i>Attention</i> × <i>Post-GME</i>	
	<i>Coefficient</i>	<i>t-stat</i>	<i>Coefficient</i>	<i>t-stat</i>	<i>Coefficient</i>	<i>t-stat</i>
[1,21]	3.87%	(1.99)	−5.53%	(−2.86)	−1.67%	(−1.66)
0	1.11%	(3.24)	−0.57%	(−1.56)	0.54%	(4.45)
1	0.29%	(1.06)	0.26%	(0.87)	0.55%	(6.83)
2	0.65%	(2.19)	−0.50%	(−1.43)	0.15%	(0.90)
3	0.33%	(2.09)	−0.41%	(−1.82)	−0.08%	(−0.67)
4	0.04%	(0.32)	0.06%	(0.27)	0.10%	(0.56)
5	0.36%	(1.98)	−0.42%	(−1.98)	−0.06%	(−0.54)
[6-10]	0.31%	(0.87)	−0.54%	(−1.40)	−0.23%	(−0.78)
[11-15]	0.39%	(0.59)	−0.93%	(−1.24)	−0.54%	(−2.28)
[16-20]	1.43%	(1.37)	−2.45%	(−2.11)	−1.02%	(−1.69)
[21-60]	−0.39%	(−0.16)	−0.69%	(−0.25)	−1.08%	(−0.70)

This table repeats the analysis in Specification 6 of Table 9 (the sample that excludes GME and AMC) after varying the return horizon. For reference, the first row reports the baseline 21-day holding period. We also report results for returns measured on the current day (i.e., day 0), each of the subsequent 5 days, the subsequent 3, and 4 weeks, and the subsequent 5 through 12 weeks. Standard errors are clustered by firm and month, and *t*-statistics are reported in parentheses.

(b) Table 10: Alternative horizons

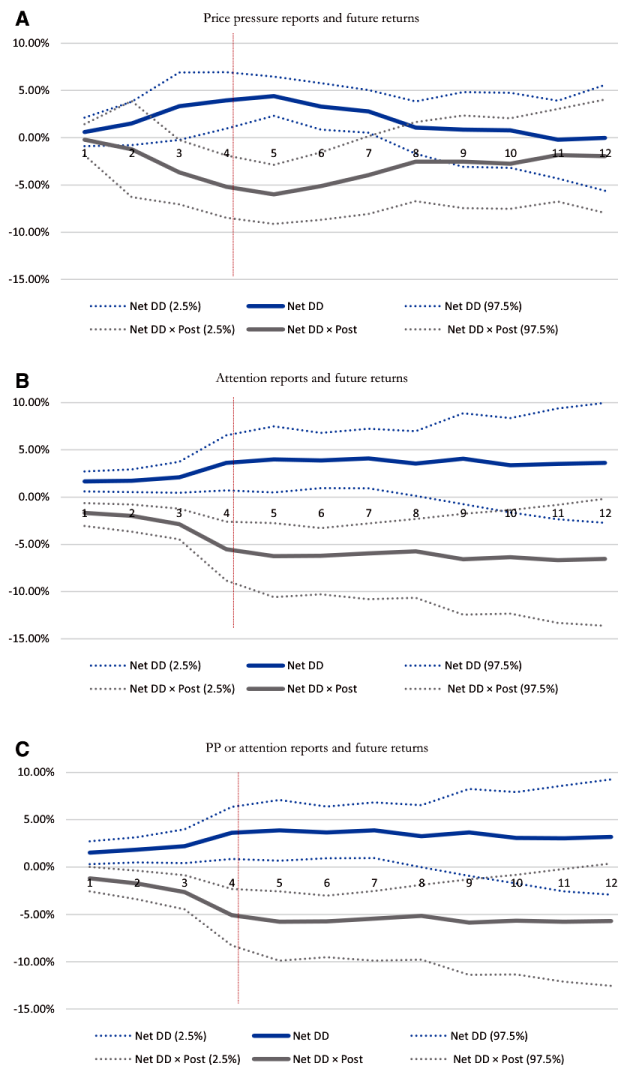


Figure 5

Figure 9: Figure 5: Price-pressure and attention reports by week

5.8 Table 11, Figure 6, and Table 12: Retail trading and investor welfare concerns

Read:

- Table 11 shows that DD reports are strongly associated with small retail order imbalances.
- Figure 6 shows the strongest effects in the extreme right tail of the small retail imbalance distribution.
- Table 12 shows that small retail imbalances following DD reports are informative before GME.
- The interaction with Post-GME is negative, indicating that the informativeness of retail trading following DD reports weakens sharply after GME.
- The effect remains partly negative after controlling for the recommendation direction, suggesting that DD report quality is not the only channel.

Economic message. WSB DD reports can coordinate retail trading. Before GME, this retail trading partly reflects information; after GME, the evidence is more consistent with uninformed or less informative trading pressure.

Table 11
WSB reports and investor order imbalances

	[1] Institutional % imbalance	[2] Large retail % imbalance	[3] Small retail % imbalance	[4] Institutional SD abn. imb.	[5] Large retail SD abn. imb.	[6] Small retail SD abn. imb.
<i>Net DD</i>	-0.19 (-0.60)	1.62 (3.82)	4.80 (6.30)	-0.67 (-1.12)	2.43 (2.86)	6.09 (6.09)
<i>Net DD × Post-GME</i>	0.08 (0.18)	-1.02 (-2.25)	0.95 (0.90)	-0.09 (-0.09)	-2.10 (-2.13)	-1.28 (-1.04)
<i>WSB nonresearch</i>	-0.22 (-1.50)	0.10 (0.66)	1.55 (3.80)	0.24 (0.64)	-1.17 (-3.12)	1.74 (3.99)
<i>WSB nonresearch × Post-GME</i>	0.32 (2.24)	-0.12 (-0.70)	-1.43 (-2.43)	0.26 (0.58)	1.21 (2.87)	-1.09 (-1.71)
<i>Net SA</i>	0.02 (0.09)	0.76 (5.53)	2.12 (9.71)	0.22 (0.88)	0.31 (1.01)	1.46 (5.14)
<i>Net SA × Post-GME</i>	0.32 (1.29)	0.22 (0.91)	0.75 (1.40)	0.15 (0.32)	0.68 (1.37)	-0.80 (-1.24)
<i>log(Size)</i>	0.94 (14.84)	0.28 (9.57)	0.22 (3.54)	0.05 (0.80)	0.28 (5.12)	0.41 (5.54)
<i>log(BM)</i>	0.01 (0.19)	-0.05 (-1.22)	-0.19 (-2.69)	0.05 (0.87)	0.06 (1.20)	0.18 (2.54)
<i>Ret [-1]</i>	-7.53 (-6.48)	10.30 (8.46)	9.03 (6.71)	-15.06 (-11.47)	16.04 (10.83)	13.43 (9.61)
<i>Ret [-5, -2]</i>	6.53 (6.07)	-5.57 (-14.28)	-8.68 (-9.14)	7.19 (6.11)	-4.38 (-6.84)	-6.90 (-7.40)
<i>Ret [-26, -6]</i>	1.12 (2.53)	-1.94 (-9.70)	-4.17 (-12.69)	0.95 (1.88)	-1.42 (-4.25)	-3.05 (-9.11)
<i>News sentiment [-1]</i>	0.31 (3.38)	0.12 (1.23)	0.11 (1.07)	0.50 (3.80)	0.16 (1.20)	0.19 (1.41)
<i>News sentiment [-5, -2]</i>	0.14 (2.88)	0.17 (3.02)	0.14 (2.08)	0.15 (2.43)	0.19 (2.56)	0.18 (1.99)
<i>News sentiment [-26, -6]</i>	-0.03 (-1.52)	0.03 (1.35)	0.01 (0.29)	-0.06 (-2.51)	0.03 (0.88)	0.00 (0.76)
<i>Abs. ret [-1]</i>	6.89 (6.12)	-5.73 (-4.24)	-3.29 (-2.53)	5.31 (4.19)	-13.30 (-6.82)	-10.05 (-6.99)
<i>Abs. ret [-5, -2]</i>	-4.04 (-3.43)	4.55 (15.61)	8.75 (13.16)	-7.77 (-5.69)	2.57 (4.18)	8.23 (14.00)
<i>Abs. ret [-26, -6]</i>	-1.49 (-3.71)	1.41 (6.07)	4.70 (11.45)	-3.06 (-6.35)	0.21 (0.64)	4.06 (8.92)
<i>Abs. news sentiment [-1]</i>	0.52 (3.80)	-0.47 (-3.14)	-0.54 (-3.40)	0.57 (3.66)	-0.88 (-4.23)	-0.71 (-4.15)
<i>Abs. news sentiment [-5, -2]</i>	0.11 (1.62)	-0.03 (-0.53)	0.04 (0.53)	0.01 (0.07)	-0.17 (-2.37)	-0.01 (-0.99)
<i>Abs. news sentiment [-26, -6]</i>	0.05 (1.68)	0.00 (0.10)	0.09 (1.88)	-0.02 (-0.42)	-0.01 (-0.32)	0.03 (0.57)
<i>Inst. % imbalance [-1]</i>	0.14 (44.52)	-0.01 (-3.01)	-0.01 (-4.25)	-0.01 (-16.10)	-0.01 (-5.43)	0.00 (0.34)
<i>Large retail % imbalance [-1]</i>	0.00 (-1.39)	0.01 (2.83)	-0.04 (-14.66)	0.00 (0.36)	-0.12 (-19.51)	-0.03 (-14.50)
<i>Small retail % imbalance [-1]</i>	-0.01 (-3.60)	0.04 (16.44)	0.12 (19.25)	0.00 (1.49)	-0.03 (-8.03)	-0.17 (-25.14)
<i>Inst. SD abn. imbalance [-1]</i>	0.02 (8.28)	0.00 (-2.75)	0.00 (2.13)	0.30 (34.70)	0.00 (0.36)	-0.01 (-1.80)
<i>Large retail SD abn. imbalance [-1]</i>	-0.01 (-6.34)	0.02 (7.82)	0.02 (9.21)	-0.01 (-5.06)	0.16 (20.49)	0.02 (6.66)
<i>Small retail SD abn. imbalance [-1]</i>	0.01 (3.34)	0.00 (0.52)	0.01 (1.77)	-0.01 (-1.79)	0.07 (19.35)	0.33 (36.89)
<i>Net DD + Net DD × Post</i>	-0.11 (-0.41)	0.60 (3.99)	5.76 (7.18)	-0.76 (-1.05)	0.33 (0.76)	4.82 (7.24)
<i>Net DD - Nonresearch</i>	0.03 (0.04)	1.52 (3.31)	3.26 (3.81)	-0.92 (-0.84)	3.61 (4.40)	4.36 (3.80)
<i>(Net DD - Nonresearch) × Post</i>	-0.24 (-0.35)	-0.90 (1.69)	2.38 (2.05)	-0.35 (-0.27)	-3.32 (-3.21)	-0.19 (-0.14)
<i>Net DD - Net SA</i>	-0.21 (-0.61)	0.87 (1.92)	2.69 (3.23)	-0.90 (-1.04)	2.13 (2.38)	4.63 (4.17)
<i>(Net DD 21 - Net SA) × Post</i>	-0.24 (-0.30)	-1.24 (-2.25)	0.21 (-0.17)	-0.24 (-0.18)	-2.79 (-2.69)	-0.47 (-0.33)

(Continued)

Table 12
Informativeness of small retail trading following WSB DD reports

	[1]	[2]	[3]	[4]
<i>Small retail imbalance</i>	0.05% (2.26)	0.05% (2.27)	0.05% (2.29)	0.05% (2.28)
<i>Small retail imbalance × Post</i>	-0.09% (-1.50)	-0.10% (-1.61)	-0.09% (-1.53)	-0.10% (-1.62)
<i>Small retail imbalance × DD</i>	1.39% (1.88)	0.54% (1.79)	0.62% (2.65)	0.35% (1.57)
<i>Small retail imbalance × DD × Post</i>	-1.50% (-1.33)	-1.11% (-3.05)	-1.01% (-1.13)	-0.76% (-2.37)
<i>Small retail imbalance × NR indicator</i>	0.80% (1.05)	0.08% (0.38)	0.09% (0.38)	0.00% (0.02)
<i>Small retail imbalance × NR indicator × Post</i>	-1.03% (-0.69)	-0.73% (-2.11)	-0.68% (-0.56)	-0.46% (-1.96)
<i>Small retail imbalance × SA</i>	0.03% (0.51)	0.02% (0.39)	0.00% (-0.04)	0.00% (0.03)
<i>Small retail imbalance × SA × Post</i>	-0.15% (-1.66)	-0.11% (-1.20)	-0.16% (-1.65)	-0.10% (-1.06)
<i>Net DD</i>			4.97% (2.46)	1.88% (2.07)
<i>Net DD × Post</i>			-4.28% (-3.20)	-2.44% (-1.68)
<i>Nonresearch</i>			4.92% (1.24)	0.53% (1.08)
<i>Nonresearch × Post</i>			-3.99% (-1.15)	-1.75% (-3.03)
<i>Net SA</i>			0.58% (2.29)	0.65% (2.64)
<i>Net SA × Post</i>			0.54% (1.33)	0.48% (1.18)
<i>Imb. × DD + Imb. × DD × Post</i>	-0.11% (-0.16)	-0.56% (-2.67)	-0.40% (-0.50)	-0.41% (-1.69)
<i>Imb. × NR indicator + Imb. × NR indicator × Post</i>	-0.23% (-0.23)	-0.65% (-2.36)	-0.59% (-0.53)	-0.46% (-2.86)
<i>Imb. × SA + Imb. × SA × Post</i>	-0.13% (-1.54)	-0.09% (-1.16)	-0.16% (-1.84)	-0.10% (-1.23)
<i>Obs. (firm-days)</i>	2,523,578	2,522,075	2,523,578	2,522,075
<i>Day FE</i>	Yes	Yes	Yes	Yes
<i>Table 3 controls</i>	Yes	Yes	Yes	Yes
<i>Include GME & AMC</i>	Yes	No	Yes	No

This table reports estimates from the following regression:

$$\begin{aligned}
 R_{i,t+1,t+21} = & \beta_1 Imb_{i,t} + \beta_2 Imb_{i,t} \times Post_t + \beta_3 Imb_{i,t} \times DD_{i,t} \\
 & + \beta_4 Imb_{i,t} \times DD_{i,t} \times Post_t + \beta_5 Imb_{i,t} \times NR\ indicator_{i,t} \\
 & + \beta_6 Imb_{i,t} \times NR\ indicator_{i,t} \times Post_t + \beta_7 Imb_{i,t} \times SA_{i,t} \\
 & + \beta_8 Imb_{i,t} \times SA_{i,t} \times Post_t + Controls_{i,t} + Day_t + \varepsilon_{i,t}.
 \end{aligned}$$

The dependent variable is the 1-month-ahead return. *Imb* is *SD abnormal small retail imbalance* (as defined in Table 11). *DD* is an indicator equal to one if a *DD* report was issued for firm *i* on day *t*. *NR indicator* and *SA* are indicators equal to one if a nonresearch post or *SA* report was issued for firm *i* on day *t*, respectively. *Controls* and *Day* are defined as in Table 3, and in the interest of brevity, the estimates on controls are unreported. We standardize all continuous variables to have mean zero and unit variance. Standard errors are clustered by firm and month, and *t*-statistics are reported in parentheses. Below the regression estimates, we report formal tests for whether certain pairs of coefficients are significantly different from zero.

(a) Table 11: Investor order imbalances

(b) Table 12: Retail trading informativeness

whether certain pairs of coefficients are significantly different from zero.

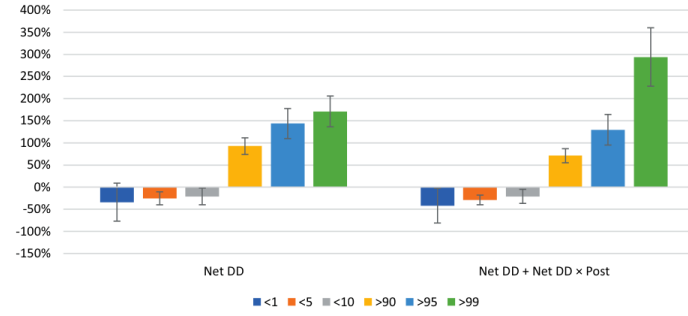


Figure 6
DD reports and small investor order imbalances: Extreme quantiles
 This figure reports estimates from Specification 6 of Table 11 after replacing *Small retail SD abnormal imbalance* with indicators equal to one if the imbalance is greater than the 90th, 95th, or 99th percentile or less than the 10th, 5th, or 1st percentile. We convert the estimates into a percentage effect by scaling the percentage point increase by the mean of the dependent variable. The figure plots the percentage estimates on *Net DD* and *Net DD + Net DD × Post-GME*. Standard errors are clustered by firm and month, and the error bars represent 95% confidence intervals.

Figure 11: Figure 6: Extreme quantiles of small retail order imbalance

6 Short summary

- The paper studies whether WSB due diligence reports contain investment value.
- It hand-collects and codes DD reports from July 2018 through June 2021.
- The sample contains 5,015 DD reports covering 906 firms.
- Pre-GME DD reports significantly predict future returns.
- The predictability remains after excluding GME and AMC.
- Future-return predictability persists over multiple weeks and does not immediately reverse.
- Pre-GME DD reports also predict cash-flow news, suggesting that some information is fundamental.
- Post-GME, DD report informativeness deteriorates sharply.
- The post-GME period features more price-pressure and high-attention reports.
- Price-pressure and attention-based reports are the ones most associated with the decline in informativeness.
- DD reports are linked to small retail order imbalances.
- Before GME, retail order imbalances following DD reports are more informative.
- After GME, the informativeness of small retail trading following DD reports disappears or becomes negative.
- Seeking Alpha reports remain a useful benchmark: their informativeness does not collapse in the same way.
- The paper's broader message is that social-media research quality can change endogenously when platform culture and participation change.

7 Conclusion

7.1 Core conclusion

Bradley, Hanousek, Jame, and Xiao show that WSB DD reports had investment value before the GameStop short squeeze but that this value deteriorated afterward. Pre-GME DD reports predict future returns and cash-flow news; post-GME DD reports increasingly emphasize price pressure and attention-grabbing stocks and no longer reliably predict positive returns.

7.2 Economic intuition

Before GME, WSB had a smaller user base and DD reports were more likely to contain analysis with firm-specific information. The GameStop event dramatically increased WSB participation and changed the platform’s culture. New contributors and readers became more attracted to strategies based on attention, short squeezes, and coordinated price pressure. Those strategies were less connected to fundamentals, so report informativeness declined.

7.3 Takeaway

The paper provides a nuanced view of retail social finance. Online communities can produce valuable investment research, but their quality is fragile. When attention and speculative coordination dominate analysis, the same platform can become less informative and potentially harmful to smaller retail investors.